

12 A MOTE OF DUST SUSPENDED IN A SUNBEAM

When I think about my father I think about books. Plural. Every time we moved, which was often, we had to make room and brought more books to the second hand bookstore than most people collect in a lifetime. He couldn't for the life of him hammer in a nail, but in his eagerness to read books, he became my Wikipedia and LLM rolled into one, long before either existed.

There is one single item that makes me think of him above all else. It's not a book but the globe I inherited from him. It went everywhere he went, and now it goes everywhere I go. For him, as for me, it signifies that we are, above all, inhabitants of planet earth. And every time I look at that globe, every time my fingers wander over the thin layer of transparent lacquer protecting the paper, I think how utterly astonishing it is that we exist. How everything that ever lived, loved and died in the known universe, did so in a layer thinner than that lacquer. And I think about that first photograph of Earth from Space. Of Carl Sagan describing it as "a mote of dust suspended in a sunbeam" ([Sagan, 1994](#)). Of how we seem mere surface tension between rock and vacuum. Isn't life miraculous? Isn't it worth protecting above all else?

In the life that led up to writing this book I've come to realize that we don't live on a candle we are burning up. Instead we inhabit a mote of dust suspended in a sunbeam.

12.1 Sunlight is Abundant but Earth Has Her Limits

The sunlight that falls upon us is abundant, but mother Earth's resources are not. Her soil is finite. Her atmosphere can only absorb so much. How close are we to the limits of mother Earth? How do we know when we've crossed them?

Jay Forrester, a professor at MIT ([MIT, 2016](#)), thought he had an approach that could give the answer. He became convinced that the human mind was excellent at identifying the components of a system (fossil fuels, people, pollution) but terrible at predicting how they interact over time through non-linear feedback loops. He argued that "mental models" were insufficient and that we needed computer simulations. Starting in the mid-1950s, he developed a field called system dynamics that models the world as a system of stocks that interact through flows, publishing his first book *Industrial Dynamics* in 1961 ([MIT, 2016](#)). I introduced system dynamics in chapter 1 and likened these stocks to bathtubs with an inflow (faucet) and outflow (drain). I jokingly called it *The Bathtub That Explains Everything*. He also build a simple world model.

System dynamics finally allowed us to model formally how abusive we had been treating mother earth. The first high impact system dynamics report was called *Limits to Growth*. It was written by Dennis and Donella Meadows, along with Jørgen Randers and William Behrens III, a team of young researchers assembled by Jay Forrester who extended his world model ([Nørgård, 2010](#)). It came out in 1972 and was an immediate blockbuster with 30 million copies sold ([Nørgård, 2010](#)). It helped that the first oil crisis began in October

1973, shortly after the book's publication ([Federal Reserve, 2013](#)), and that it was actively pushed to world leaders and the media by a group consisting primarily of European industrialists and scientists known as *The Club of Rome* ([Club of Rome, 2021](#)). But I think it's success was mostly because it proved mathematically what most of us had felt intuitively.

12.2 Doughnut Economics: the Floor and the Ceiling

This book makes a claim that might seem contradictory: we can have abundance *and* stay within planetary limits. Most frameworks force you to choose. Environmentalists warn the limits force us to choose austerity. Economists promise endless growth. Each camp suspects the other of dangerous naivety. But I hope this book shows the math points somewhere neither camp expected.

The economist Kate Raworth became weary of both camps and came up with a synthesis. She captured the two requirements brilliantly with her image of the doughnut ([Raworth, 2012](#)). Do you see a doughnut in your mind? Now imagine that the outer edge denotes earth's ecological ceiling: the planetary boundaries we cannot transgress without destabilizing Earth's systems. Further imagine the inner edge is the social foundation: the minimum level of human welfare below which dignity fails. Food, water, shelter, health, energy, education, political voice. Kate Raworth based the inner edge on twelve dimensions of human wellbeing derived from the United Nations Sustainable Development Goals ([Raworth, 2017](#)). Right now, there are billions of people for whom the minimum level of dignity is not met ([UN, 2024](#)).

The doughnut itself, the ring of actual pastry, is the safe and just space for humanity. The sweet spot.

Most discussions of the doughnut treat it as aspirational, a distant goal we might approach if the rich sacrifice enough. Raworth herself tends toward caution about growth. She would probably raise an eyebrow at my optimism about providing abundance for everyone within planetary boundaries. But I think she might also be pleasantly surprised by where the numbers actually lead.

In the first chapter of the book I explained how the learning curve changes everything. When solar electricity costs are a tenth of what they were a decade ago and continue falling, when batteries follow the same trajectory, when the material intensity of energy production drops by orders of magnitude, the doughnut becomes more than aspirational. Abundance becomes achievable *without* transgressing boundaries, precisely because we're replacing high-impact systems with low-impact ones. The trade-off that haunted twentieth-century environmentalism, prosperity versus planetary health, dissolves.

I don't say this lightly. In chapter 1.6 I confessed my Calvinist roots and my affinity to degrowth as a way to explain to my guilty conscience why I have it so good when so many haven't. But the numbers don't lie. If we deploy solar, wind, and batteries at the rates the learning curves enable, we can provide abundant energy to every human on Earth while

staying within all nine planetary boundaries. Not by restricting demand. Not by immiserating the poor. By choosing technologies that are simply better on both dimensions.

12.3 Abundance Within Planetary Boundaries

This brings us to the framework we have been circling around in recent chapters without naming it. Chapter 1.6 examined what we take from the Earth: the raw materials, the mining, the extraction, the inflow to our planetary bathtub. Chapter 8 examined the bathtub itself: the land, the finite surface, the competing uses. Chapter 1.9 examined what we put back: the pollution, the emissions, the seven million deaths, the outflow. Inflow, bathtub, outflow: system dynamics applied to the planet.

In 2009, Johan Rockström and his colleagues gave the ceiling of Raworth's doughnut precise coordinates ([Rockström, 2009](#)). They identified nine processes that regulate Earth's stability and measured how close we are to destabilizing each one. Updated by Katherine Richardson in 2023 ([Richardson, 2023](#)), the framework is essentially a dashboard for Spaceship Earth. Where Raworth shows us *why* boundaries matter (human flourishing requires both a floor and a ceiling), Rockström shows us *where* those boundaries lie and how far we've transgressed them.

The standard approach to planetary boundaries is to present all nine, show we've crossed most, and leave the reader either paralyzed or numbly indifferent. A bit like the teacher who told me we were burning up the Earth like a candle and then offered no solution. I refuse to do that to you. Instead, let me use the bathtub metaphor to show that we can solve these problems with an energy system that provides abundant energy to everyone while staying within the outer boundary of the doughnut.

Let's define the earth's ecosystem as the bathtub. By the earth's ecosystem I mean that thin layer around the globe consisting of fertile soil, breathable atmosphere, and the living beings that inhabit them. If you look at the planetary boundaries framework you will see that most boundaries are about what we are dumping into that bathtub. Climate change. Ocean acidification. Ozone depletion. Aerosol loading. Biogeochemical flows of nitrogen and phosphorus. Novel entities. Commoner's law returns: everything must go somewhere. The planetary boundaries are largely a catalogue of where we've been putting things we shouldn't. Now let's see how we solve it.

SIDEBAR: The Nine Planetary Boundaries (2023 Status)

Boundary	Status	Category	Primary Solutions
Stratospheric ozone	Healing	Inflow	Already solved (Montreal Protocol)
Climate change	Transgressed	Inflow	Solar + wind + food reform
Ocean acidification	Approaching	Inflow	Solar + wind + food reform

Boundary	Status	Category	Primary Solutions
Atmospheric aerosol loading	Regional issues	Inflow	Solar + wind + food reform
Biogeochemical flows (N, P)	High risk	Inflow	Precision agriculture + meat replacements
Novel entities	Transgressed	Inflow	Pesticide reduction + plastics regulation
Land-system change	Transgressed	Bathtub	Meat replacements (free up land)
Freshwater change	Transgressed	Bathtub	Meat replacements + precision irrigation
Biosphere integrity	High risk	Bathtub	Meat replacements (free up land)

Key: Within safe operating space | Transgressed (increasing risk) | Highly transgressed (high risk)

Source: ([Richardson, 2023](#))

The sidebar categorizes each boundary as "inflow" or "bathtub". The same distinction we've been using since Chapter 1.1. The pattern holds all the way up to the planetary scale.

Solar and Wind Solve a Lot

Three inflow boundaries share a common pattern: they are mainly, for about 75%, caused by burning things ([WRI, 2024](#)). I'm talking about climate change, ocean acidification, and aerosol loading.

Coal, oil, and gas release carbon dioxide that warms the climate and acidifies the oceans when absorbed. Combustion also releases particulate matter that kills seven million people every year ([WHO, 2024](#)).

The solution is obvious. Stop burning biomass and produce the bountiful energy provided by father sun directly with solar panels and wind turbines. They don't emit anything into the atmosphere.

Fix Agriculture and All the Inflow Boundaries are Solved

Once we stop burning biomass, the rest of the inflow for these three boundaries (climate change, ocean acidification, and aerosol loading) comes from agriculture ([WRI, 2024](#)). Through land use change (burning or cutting down forests) and burping cows and sheep.

On top of that agriculture causes the transgression of the nitrogen and phosphorus cycles. They do that by using more fertilizer than crops can absorb so that the rest runs off and wreaks havoc with ecosystems. ([Mongabay, 2021](#)) Agriculture is also a significant contributor to the planetary boundary called novel entities ([Campbell, 2017](#)). This is mainly about pesticides that kill far more insects, insect eaters (like birds), and plants than is sustainable.

The solution is two fold. Meat replacements would reduce all agricultural problems by about two thirds, because animals are a very inefficient way to produce food ([Poore, 2018](#)) (see chapter 1.8). We are talking about plant-based replacement, precision fermentation, or cultivated meat. We would still have the same amount of food and protein, but the use of land, fertilizer and pesticide would be drastically reduced. Chapter 1.8 contains more details.

The second solution is precision agriculture: apply fertilizer more precisely (so runoff can be up to five times less) ([EUNEP, 2016](#)) ([TNC, 2017](#)) and replace pesticides with weeding robots and crops that are strategically combined to support natural predators of insects that harm crops.

So we already solved one inflow boundary: ozone. Transitioning to solar, wind and smarter agriculture can almost completely solve the other five. What's left?

Agriculture Can Also Fix The Bathtub Itself

What's left is the three boundaries that are about the bathtub itself: land-system change, freshwater change, and biosphere integrity. These have little to do with energy for our machines. Unless you're foolish enough to pursue bioenergy at scale. What affects them is, again, how we produce energy for our bodies.

We already discussed the enormous land use of agriculture (especially for meat) in paragraph 11.3.

Freshwater change tracks our disruption of the water cycle. Agriculture accounts for roughly 70 percent of global freshwater withdrawals ([FAO, 2024](#)). Aquifers drain faster than they recharge ([Jasechko, 2024](#)).

Biosphere integrity is perhaps the gravest planetary boundary we've crossed. Extinction rates run 10 to 100 times the natural background ([Ceballos, 2015](#)). We are living through the sixth mass extinction ([Ceballos, 2015](#)). The main cause is the land use by agriculture, followed by pesticide use and fertilizer runoff by agriculture ([UNEP, 2021](#)). Since meat is responsible for about 80 percent of agricultural land use ([Ritchie, 2024](#)) and indirectly for most pesticide and fertilizer use, meat replacements are the biggest lever we have to stop the 6th mass extinction.

Fortunately, I think that meat replacements will show a steep learning curve. They simply need fewer materials and cause fewer problems (still aside from the animal welfare issue). I'm convinced they will become a no-brainer once they slid down the learning curve enough to become cheaper, healthier and tastier.

Then we still need to stem the inflow of plastics, that portion of novel entities that isn't pesticides. We still need to improve some industrial processes like cement, and steel. We need a way to run long distance aviation and shipping on electrically produced fuels. But abundant energy within planetary boundaries is not the problem.

12.4 Silver Bullets

There is a phrase that makes me want to throw things: 'There are no silver bullets.' I've noticed it gets deployed whenever someone proposes a solution that actually works. It sounds wise but it counsels paralysis. 'Find me some goddamn bullets' would be more useful.

Solar panels. Wind turbines. Batteries. Digital power converters. And for the hard cases, hydrogen or synthetic fuels made from electricity. That's basically all we need to stop burning nature to feed our machines. Five technologies, all riding learning curves, all getting cheaper every year, all already deployed at scale somewhere in the world.

For agriculture, which drives most of the rest of the ways we transgress boundaries, you need meat replacements and precision farming, both of which are riding learning curves.

I used to look at that globe and feel wonder tinged with guilt over what we were doing to it. By pictures of the earth burning up. No longer. These days I enjoy the sight and feel wholeheartedly. Because now I understand we can all flourish while protecting this mote of dust suspended in a sunbeam. Isn't it marvellous?